

PERSONAL INFORMATION



Shubham Singh

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SUMMARY

I am a Postdoctoral Researcher in the Digital Twin Research Unit, Institute of Integrated Research, Institute of Science Tokyo (Japan), working with Prof. Katsuki Fujisawa on safe autonomous systems with reinforcement learning. I received my Ph.D. in Control Systems and Automation from the Indian Institute of Technology Jodhpur under the supervision of Prof. Anoop Jain, following a B.Tech. and an M.Tech. in Electrical Engineering. My research addresses safety-critical control, learning-based decision-making and multi-agent robotic systems, with expertise in Barrier Lyapunov Functions (BLFs), Control Barrier and Control Lyapunov Functions (CBF/CLF), reinforcement learning and POMDP-based planning under uncertainty. These methods are validated experimentally on aerial (Crazyflie 2.1) and ground (Khepera IV) robots and published in venues including IEEE Transactions on Automatic Control and Systems & Control Letters.

WORK EXPERIENCE

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| 09/2026 – current | <p>Postdoctoral Researcher — Digital Twin Research Unit, Institute of Integrated Research, Institute of Science Tokyo, Japan</p> <ul style="list-style-type: none">• Research on safe autonomous systems with reinforcement learning, hosted by Prof. Katsuki Fujisawa. |
| 01/2026 – 06/2026 | <p>Research Associate — DSC Lab, Department of Electrical Engineering, IIT Jodhpur, India</p> <ul style="list-style-type: none">• Conducted independent research on safe control under uncertainty and obstacle-cluttered environments using optimization-based and reinforcement learning techniques.• Mentored junior Ph.D. students and contributed to the lab's ongoing funded projects. |
| 07/2020 – 12/2025 | <p>Ph.D. Research Scholar — DSC Lab, Department of Electrical Engineering, IIT Jodhpur, India</p> <ul style="list-style-type: none">• Developed the Möbius-BLF framework for safety-critical control of unicycle robots within nonconcentric circular geofences.• Extended the framework to multi-agent coordination under spatio-temporal flow fields; hardware-validated on Crazyflie 2.1 UAVs and Khepera IV UGVs. |
| 31/07/2020 – 30/06/2025 | <p>Teaching Assistant — IIT Jodhpur, India</p> <ul style="list-style-type: none">• Control Systems; Mobile Robots; Introduction to Cyber-Physical Systems; Security in CPS; Signals and Systems; Basic Electrical Engineering; Electrical Machines Lab; Digital Design (Verilog HDL). |

EDUCATION AND TRAINING

- 31/07/2020 – 06/06/2026** **Ph.D. in Control Systems and Automation**, Indian Institute of Technology Jodhpur, India — *First Class*
Thesis: Coordination and Control of Robotic Systems under Nonuniform Motion Constraints
Supervisor: Prof. Anoop Jain
- 01/07/2018 – 15/06/2020** **M.Tech. in Electrical Engineering**, National Institute of Technology Hamirpur, India
Thesis: Synchronization of Chaotic System Using Adaptive Backstepping Control — First Class
Supervisor: Prof. Himesh Handa
- 01/07/2014 – 01/06/2018** **B.Tech. in Electrical Engineering**, Bhilai Institute of Technology, Durg, India
Thesis: Speed Control of Induction Motors — First Class
Supervisor: Prof. Shraddha Kaushik

SELECTED TALKS

- 12/06/2024** *Constrained Synchronization of Drive and Response Chaotic Systems with Parametric Uncertainty using Barrier Lyapunov Function*
American Control Conference (ACC) 2024.
- 09/06/2022** *Time-Varying Distance-Constrained Formation Maneuver Control with Guaranteed Prescribed Performance for Mobile Agents*
American Control Conference (ACC) 2022.
- 22/12/2021** *Distance-Constrained Formation Control of Multi-Agent Systems Using Asymmetric Barrier Lyapunov Function*
Seventh Indian Control Conference (ICC) 2021 — **Best Student Paper Award Finalist.**

PEER REVIEWING ACTIVITY

- Ongoing** **Journals:** IEEE Transactions on Automatic Control (TAC); IEEE Control Systems Letters (L-CSS); IEEE Transactions on Cybernetics; Nonlinear Dynamics; IEEE Transactions on Network Science and Engineering; Springer Nature (Scientific Reports)
Conferences: Conference on Decision and Control (CDC); 22nd IFAC Symposium on Automatic Control in Aerospace (ACA 2022); Indian Control Conference 2025; Advances in Robotics 2025.

RESEARCH PROJECTS

- 01/2026 – 06/2026** **Safe Control under Uncertainty and Obstacle Avoidance via Optimization and RL** — IIT Jodhpur (Research Associate)
 - Developed a unified framework combining RL and optimization-based control for safe navigation in uncertain, obstacle-cluttered environments.

- Integrated CBFs and CLFs within a QP-based controller for real-time safety and stability guarantees.
- Investigated PPO-based learning policies for adaptive circumnavigation under bounded disturbances and unknown dynamic obstacles.

11/2025 – 06/2026

RL-Driven Detection and Mitigation of False Data Injection in Formation Control — IIT Jodhpur, DSC Lab

- Developed a secure distance-constrained formation control framework under inter-agent False Data Injection (FDI) attacks.
- Designed RL-based real-time attack detection and optimization-driven defense mechanisms for resilient multi-agent formations.

01/2026 – 06/2026

Optimization-Based Robust Circumnavigation Control via CLF–CBF — IIT Jodhpur, DSC Lab

- Developed a QP-based multi-agent circumnavigation controller integrating CLF and CBF under external disturbances; validated via MATLAB simulations.

07/2023 – 12/2025

Nonuniform Motion Constraint in Circular Geofencing — IIT Jodhpur, DSC Lab (Ph.D. thesis core)

- Developed the Möbius-BLF framework for nonconcentric circular geofencing; validated on Crazyflie 2.1 quadrotors and Khepera IV ground robots.
- Extended to multi-agent phase-coordinated formation control under spatio-temporal flow fields.

07/2021 – 06/2023

Distance-Based Formation Control with Collision Avoidance — IIT Jodhpur, SERB Project

- Proposed distributed control using time-varying asymmetric BLFs, ensuring formation with collision avoidance and connectivity preservation.

01/2022 – 03/2023

Bounded Voltage Regulation in DC Microgrids — IIT Jodhpur Research Project

- Proposed distributed voltage regulation with prescribed transient bounds; designed a load current estimator validated through MATLAB simulations.

AWARDS AND HONOURS

2021 — Best Student Paper Award Finalist, Indian Control Conference

2018, 2019 — GATE Qualified, Electrical Engineering

2026 — Selected for IPDF at IIT Bombay and IIT Delhi

PUBLICATIONS

Journal Articles

- [1] **S. Singh** and A. Jain, “Nonconcentric Boundary-Constrained Circular Formation Control for Multi-Agent Systems in the Presence of Spatiotemporal Flow Fields,” *ASME Journal of Mechanisms and Robotics*, 2026. DOI: 10.1115/1.4072559
- [2] **S. Singh** and A. Jain, “Möbius-Transformation-Based Circular Motion Control for Unicycle Robots in Nonconcentric Circular Geofences,” *IEEE Transactions on Automatic Control*, vol. 71, no. 5, pp. 2918–2933, 2026. DOI: 10.1109/TAC.2025.3631525

- [3] **S. Singh** and A. Jain, “Collision Avoidance and Connectivity Preservation using Asymmetric Barrier Lyapunov Function with Time-Varying Distance-Constraints,” *Systems & Control Letters*, vol. 183, 105672, 2024. DOI: 10.1016/j.sysconle.2023.105672
- [4] **S. Singh**, V. Vaishnav, A. Jain and D. Sharma, “Bounded Voltage Regulation in a Direct Current Microgrid using Barrier Lyapunov Function with Uncertain Load Current,” *IEEE Control Systems Letters*, vol. 7, pp. 991–996, 2023. DOI: 10.1109/LCSYS.2022.3230088
- [5] **S. Singh** and A. Jain, “Phase-Coordinated Multi-Agent Circular Formation Control with Non-Concentric Boundary Constraints,” *IEEE Transactions on Automatic Control* (conditionally accepted as a full paper). arXiv:2509.08534 [Accepted in IEEE Transactions on Automatic Control as full paper.]

Conference Papers

- [5] A. Singh, **S. Singh** and A. Tiwari, “Enforcing a Strict Safety Margin in LLM Text Generation via Barrier Lyapunov Control,” *Indian Control Conference (ICC)*, 2026 (accepted).
- [6] **S. Singh***, A. Tiwari* and A. Singh, “Safe Unicycle Navigation Among Multiple Obstacles Under Bounded Disturbances: A Learning-Augmented CLF-CBF Approach,” *Indian Control Conference (ICC)*, 2026 (accepted).
- [7] **S. Singh** and A. Jain, “Constrained Synchronization of Drive and Response Chaotic Systems with Parametric Uncertainty using Barrier Lyapunov Function,” *American Control Conference (ACC)*, pp. 2476–2481, 2024. DOI: 10.23919/ACC60939.2024.10644552
- [8] **S. Singh** and A. Jain, “Time-Varying Distance-Constrained Formation Maneuver Control with Guaranteed Prescribed Performance for Mobile Agents,” *American Control Conference (ACC)*, pp. 5134–5139, 2022. DOI: 10.23919/ACC53348.2022.9867847
- [9] **S. Singh** and A. Jain, “Distance-Constrained Formation Control of Multi-Agent Systems Using Asymmetric Barrier Lyapunov Function,” *Seventh Indian Control Conference (ICC)*, pp. 395–399, 2021. Best Student Paper Award Finalist. DOI: 10.1109/ICC54714.2021.9703147
- [10] **S. Singh** and H. Handa, “Synchronization of MLS Chaotic System Using Adaptive Backstepping Control,” *7th International Conference on Signal Processing and Integrated Networks (SPIN)*, 2020. DOI: 10.1109/SPIN48934.2020.9070983
- [11] **S. Singh**, A. Tiwari and H. Handa, “Synchronization of Hyperchaotic Rossler System using Backstepping and Sliding Mode Control,” *2nd International Conference on Innovative Mechanisms for Industry Applications (ICIMIA)*, 2020. DOI: 10.1109/ICIMIA48430.2020.9074907

* Equal contribution.

IT SKILLS

Programming Languages: MATLAB/Simulink, Python, C++ (basic), Verilog

Tools & Frameworks: ROS / ROS2, Gazebo, LaTeX, Ipe, FPGA (Verilog HDL)

Robotics Platforms: Crazyflie 2.1 (UAV), Khepera IV (UGV), TurtleBot3

Control & Optimization: Nonlinear control design, CBF/CLF-QP, Reinforcement Learning (PPO, actor-critic), POMDP